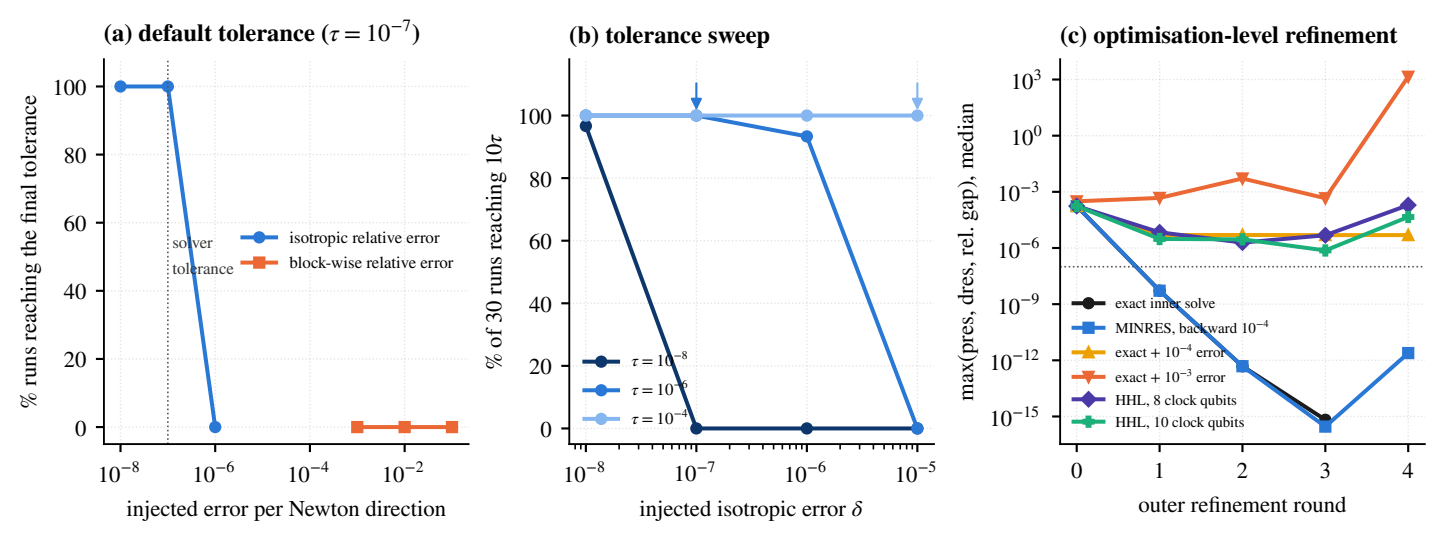
(a) default tolerance ($\tau = 10^{-7}$)
% runs reaching the final tolerance
100
80
60
40
20
0
solver
tolerance
isotropic relative error
block-wise relative error
10^{-8}
10^{-6}
10^{-4}
10^{-2}
injected error per Newton direction
(b) tolerance sweep
% of 30 runs reaching 10τ
100
80
60
40
20
0
$\tau = 10^{-8}$
$\tau = 10^{-6}$
$\tau = 10^{-4}$
10^{-8}
10^{-7}
10^{-6}
10^{-5}
injected isotropic error δ
(c) optimisation-level refinement
max(pres, dres, rel. gap), median
10^{3}
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
exact inner solve
MINRES, backward 10^{-4}
exact + 10^{-4} error
exact + 10^{-3} error
HHL, 8 clock qubits
HHL, 10 clock qubits
0
1
2
3
4
outer refinement round